

PHYS 570/MA 595: Unit 1 Exercises, Due September 11, 2025

Sometimes important definitions, derivations, and examples will be deferred to the exercises, but they are not meant to be hard. Also, the exercises will necessarily skew towards math in the beginning of this course. Exercises will be added to this list but no new problems will be added after Thursday September 4.

Thursday August 28

1. The **Pauli matrices** X , Y , and Z are defined by

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Show that they generate a group of order 16 under matrix multiplication.

2. **Draw** the lattice of subgroups for the dihedral group D_8 of order 8.

Tuesday September 2

3. A sequence of group homomorphisms

$$\cdots G_{i-1} \xrightarrow{\phi_{i-1}} G_i \xrightarrow{\phi_i} G_{i+1} \cdots$$

is called an **exact sequence** if $\text{Im}(\phi_{i-1}) = \text{Ker}(\phi_i)$ for all i . Let 1 denote the trivial group. An exact sequence of the form

$$1 \longrightarrow A \xrightarrow{\psi} B \xrightarrow{\phi} C \longrightarrow 1$$

is called a **short exact sequence**, and we call the group B an **extension** of C by A .

Show that ψ is injective and ϕ is surjective.

4. Let H and K be groups and $\phi : K \rightarrow \text{Aut}(H)$ a homomorphism. Let G be the set of ordered pairs (h, k) where $h \in H$ and $k \in K$.

Check that the binary operation

$$(h_1, k_1) \cdot (h_2, k_2) = (h_1(k_1 \triangleright h_2), k_1 k_2)$$

defines a group structure on the set $H \times K$, where $\triangleright$ is the action of K on H determined by ϕ . The resulting group is called the **semidirect product** of H and K and is written $H \rtimes_{\phi} K$ or $H \rtimes K$ when the action of K on H is clear from context.

5. **Explain** why $B_2 \cong \mathbb{Z}$, where B_2 is the 2-strand braid group.
6. Recall the definition of an isomorphism in a category. **Show** that a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ maps isomorphisms to isomorphisms.

(Exercises continued on the next page.)

Thursday September 4

7. Suppose P_i is a square matrix of fixed dimension with $P_i^2 = I$ for $i = 1, 2, \dots, n$ and suppose further that $P_i P_j = P_j P_i$ for all $i, j = 1, 2, \dots, n$. Here I is the identity matrix. Let H be a self-adjoint (Hermitian) matrix which can be written as

$$H = \sum_{i=1}^n \frac{1}{2}(1 - P_i),$$

so that H can be interpreted as a **commuting projector Hamiltonian**, and that we are guaranteed that eigenstates of H with eigenvalue 0 are given by the common eigenstates of the P_i with eigenvalue +1.

Use that commuting matrices are simultaneously diagonalizable to argue that the smallest nonzero eigenvalue of H is at least 1.